MANUKURA

Year 10 Science Practice Papers

End of Year Examination Preparation

Two papers • 45 minutes each • 100 marks

Name: _____ Class: _____

Instructions: Attempt every question. Show working for calculations. Include units. Use clear scientific vocabulary. Diagrams should be labelled with straight lines or arrows.

These papers practise the knowledge and skills assessed across the course. The situations, numbers and substances are different from the secure examination. Complete Paper A without notes, mark it, revise gaps, then complete Paper B.

Paper A overview

Section	Learning area	Marks
A	Forces and Motion	13
B	Chemistry	13
C	Body Systems	14
D	Science Skills	10
	Total	50

Formula sheet

Quantity	Equation	Unit
Speed	speed = distance ÷ time	m/s
Acceleration	acceleration = (final speed - initial speed) ÷ time	m/s^2
Force	force = mass × acceleration	N
Work	work = force × distance	J

Paper A Forces and Motion 13 marks

A1 A shopping trolley has a 95 N push forward and 38 N friction backward. Calculate the net force and state its direction. [3]

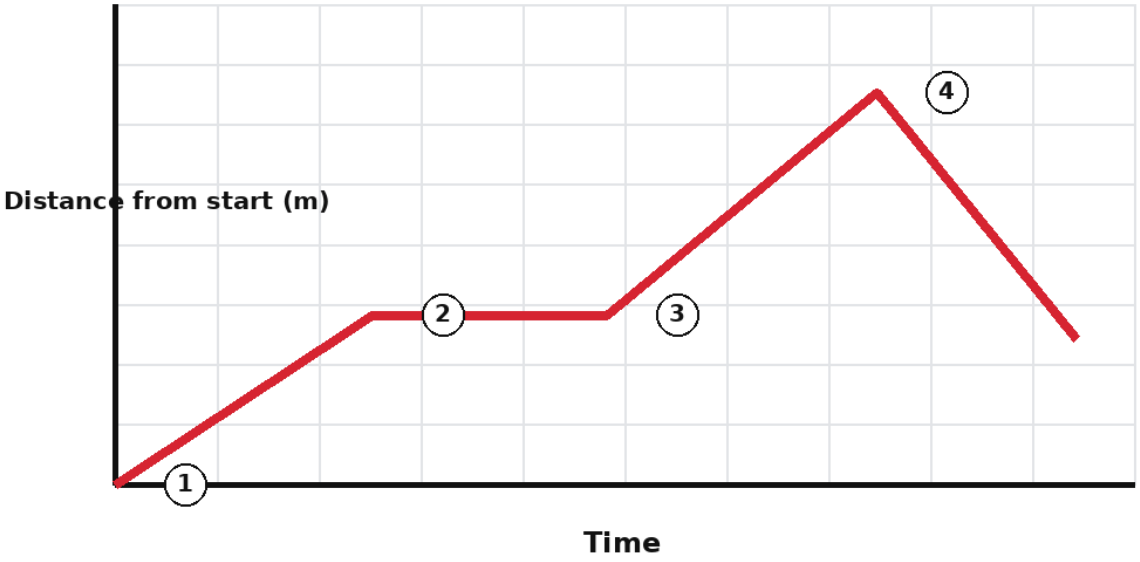
_____

_____

A2 A cyclist travels 240 m in 40 s. Calculate the average speed. [3]

_____

_____



A3 For the distance-time graph above: identify the stationary section, the fastest outward section, and the returning section. Explain how the gradient supports one answer. [4]

_____

_____

_____

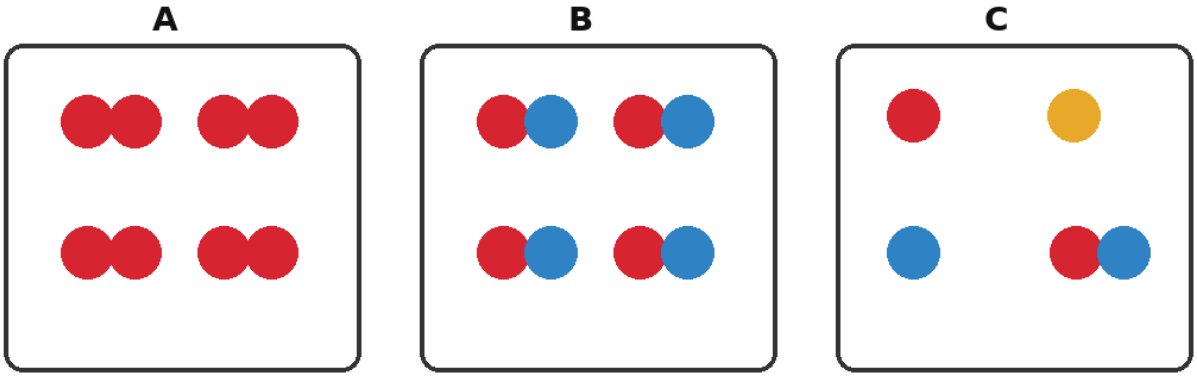
A4 A swimmer pushes water backward. Explain the action-reaction force pair using Newton's Third Law. [3]

_____

_____

_____

Paper A Chemistry 13 marks



A5 Classify particle diagrams A, B and C as element, compound or mixture. Give one reason for diagram B. [4]

_____

_____

_____

A6 A sodium atom has 11 protons, 12 neutrons and 11 electrons. State its atomic number, mass number and electron arrangement. [4]

_____

_____

_____

A7 A solution has pH 10. Classify it and predict its universal-indicator colour. [2]

_____

_____

A8 Describe how to separate iron filings, sand and salt. Explain the physical property used at each step. [3]

_____

_____

_____

_____

Paper A Body Systems 14 marks

A9 Explain elbow flexion using biceps, triceps, contracts and relaxes. [4]

_____

_____

_____

_____

A10 State the function of each: tendon, ligament, cartilage and bone marrow. [4]

_____

_____

_____

_____

A11 Describe what happens to the diaphragm, chest volume and air pressure during inhalation. Then state the direction oxygen diffuses at the alveoli. [6]

_____

_____

_____

_____

_____

_____

Paper A Science Skills 10 marks

A12 A class changes ramp height and measures the time a trolley takes to travel 2 m. Identify the independent variable, dependent variable and two controlled variables. [4]

_____

_____

_____

_____

Ramp height cm	Trial 1 s	Trial 2 s	Trial 3 s
10	3.2	3.0	3.1
20	2.7	2.5	2.6
30	2.1	2.0	2.2

A13 Calculate the mean time for the 30 cm ramp and describe the pattern in the data. [3]

_____

_____

_____

A14 Give one limitation of using a hand-held stopwatch and one specific improvement. [3]

_____

_____

_____

Paper B

45 minutes • 50 marks

Name: _____ Class: _____

Formula sheet

Quantity	Equation	Unit
Speed	speed = distance ÷ time	m/s
Acceleration	acceleration = (final speed - initial speed) ÷ time	m/s^2
Force	force = mass × acceleration	N
Work	work = force × distance	J

Paper B Forces and Motion 13 marks

B1 Draw and label a free-body diagram for a book resting on the floor of a bus moving at constant speed. Explain why the book has zero acceleration. [3]

_____

_____

_____

_____

B2 A runner increases speed from 2 m/s to 14 m/s in 6 s. Calculate acceleration. [3]

_____

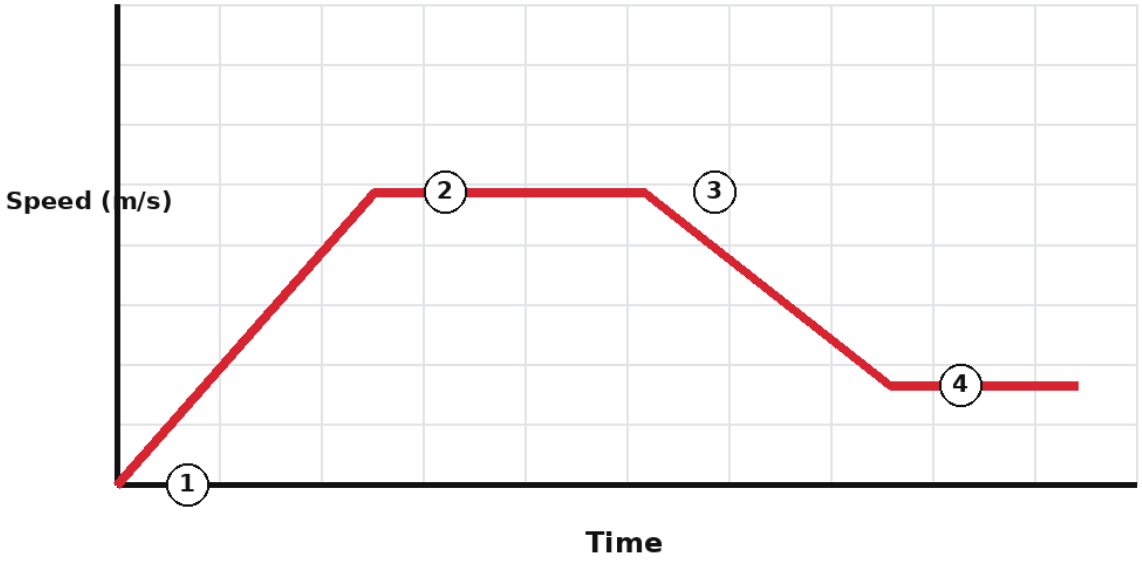
_____

B3 A 6 kg medicine ball accelerates at 3 m/s². Calculate the net force. If that force moves the ball 4 m, calculate the work done. [4]

_____

_____

_____



B4 Identify one accelerating section, one constant-speed section and one decelerating section on the speed-time graph. [3]

_____

_____

_____

Paper B Chemistry 13 marks

B5 Water vapour forms droplets on a cold window. Name the change of state and explain it using particle energy and movement. [3]

_____

_____

_____

B6 Magnesium has atomic number 12. State its proton number and electron arrangement, then predict the ion charge it forms. [4]

_____

_____

_____

B7 Magnesium reacts with dilute acid. Give two observations that show a reaction occurred and describe the test for the gas produced. [3]

_____

_____

_____

B8 Explain why a chromatography start line is drawn in pencil and why the sample spots begin above the solvent. [3]

_____

_____

_____

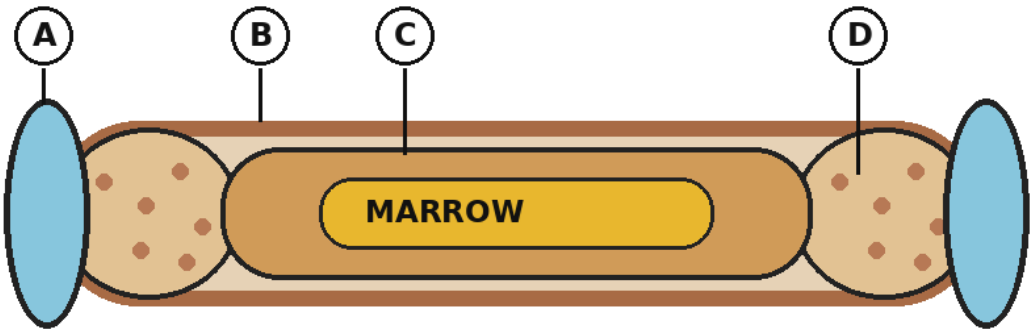
Paper B Body Systems 14 marks

B9 Name the three muscle types and give one location for each. [3]

_____

_____

_____



Identify A-D using the word bank provided.

B10 Use the word bank cartilage, compact bone, periosteum, spongy bone to identify A-D. Choose one structure and explain its function. [4]

_____

_____

_____

_____

B11 Explain how the quadriceps and hamstrings work as an antagonistic pair to extend the knee. [3]

_____

_____

_____

B12 Explain two adaptations of alveoli and how each increases the rate of gas exchange. [4]

_____

_____

_____

_____

_____

Paper B Science Skills 10 marks

B13 A student investigates whether exercise time affects breathing rate. Write a testable aim, identify the independent and dependent variables, and name one control. [4]

_____

_____

_____

_____

_____

Exercise time min	Breaths per minute
0	15
1	24
2	31
3	37
4	41

B14 Write an appropriate graph title, state the x-axis and y-axis labels, and describe the pattern. [3]

_____

_____

_____

_____

B15 Give one conclusion supported by the data and one improvement that would make the evidence more reliable. [3]

_____

_____

_____

_____